Saleha Zainab Fatima

Islamabad, Pakistan
+923185319226 | saleha.zf22@gmail.com | linkedin.com/in/saleha-zainab-fatima | github.com/saleha-zf

Summary

Software Engineer specializing in **Multimodal Machine Learning and Data Fusion**. I build end-to-end pipelines integrating **imagery, time-series, telemetry, and multi-sensor data** for real-world deployments. Experienced in full AI development cycle: from raw data preprocessing and feature engineering to deploying trained models and maintaining documentation.

Education

BE Software Engineering
National University of Sciences & Technology (NUST), Islamabad 2022-2026

Experience

Team Lead Data & AI | AI & GIS Engineer *On-site*
GreenAI, National Science and Technology Park (NASTP) June 2025 - Ongoing

- Lead the Data & AI team owning model design, data pipelines, and data repository standards from development to deployment.
- Built multi-modal crop classification models using time-series data to identify major Pakistani Kharif/Rabi crops with 87%+ accuracy, then built a dasymetric yield estimation pipeline fusing Sentinel-1 SAR and Sentinel-2 optical data to forecast district and farm-level yields before harvest.
- Built a flood monitoring microservice using Sentinel-1 SAR to classify flood, seasonal, and permanent water per district weekly, validated against UNOSAT and Copernicus EMS flood maps.
- Built a pixel-level labeling and auto-retraining AI dashboard used by field agents, data collectors, data validators, and AI developers for annotation, validation, and model versioning.
- Developed a multimodal pipeline that transforms satellite data into natural language diagnostics by integrating spatial feature extraction with RAG-based LLM workflows.

Deep Learning Research Intern *On-site*
Machine Vision and Intelligent Systems Lab Feb 2025 - June 2025

- Published research in IEEE Xplore on Semi-Supervised Contrastive Learning, developing methods to extract features from high-dimensional, unlabeled visual datasets. (DOI: 10.1109/ICET66147.2025.11321433)
- Designed annotation-efficient pipelines using Self-Supervised Learning (SSL) to analyze temporal sequences and learn embeddings, reducing manual labeling dependency while maintaining accuracy in data-scarce environments.

Projects

National-Scale Crop Classification Models | *PyTorch, TensorFlow, GDAL, Google Earth Engine, Docker, FastAPI*

- Developed multi-modal crop classification models fusing satellite and meteorological data for major Pakistani crops.
- Implemented parallel processing using multiple workers on a single server to handle large-scale satellite data.
- Combined LSTM architecture with gradient boosting ensembles to improve prediction accuracy across seasons.
- Validated with 87%+ accuracy and deployed as a microservice on government servers for an upcoming pilot project.

Weekly Flood Monitoring Microservice | *Python, Google Earth Engine, RabbitMQ, Docker, Kubernetes, rasterio*

- Built a per-district flood detection service using Sentinel-1 SAR and accumulated rainfall, separating flood, seasonal, and permanent water with orbit-matched reference stacks and bimodality-gated thresholding.
- Validated against UNOSAT and Copernicus EMS flood footprints; deployed as a containerized microservice on Kubernetes.

UAV Flight Analysis & Telemetry Visualization Engine | *Flask API, Node.js, Leaflet.js, ArduPilot, QGIS, PostgreSQL*

- Designed a high-performance engine to parse raw UAV telemetry and flight logs into structured PostgreSQL formats.
- Implemented real-time geospatial analytics to compute mission-critical metrics of UAV flights.
- Developed an interactive web-based GIS dashboard using Leaflet.js to visualize 3D motion parameters and flight paths.

Multimodal Phenology & Yield Forecasting Framework (FYP) | *PyTorch, Rasterio, Dask, Docker, FastAPI, MERN stack*

- Developed a sensor-agnostic framework for crop phenology and yield estimation that adapts to varying image modalities (RGB/Multispectral) and resolutions, enabling deployment across mixed-sensor drones without retraining.
- Deployed an imagery processing system that fuses available data sources into near real-time crop health indicators, integrated with a RAG-based LLM to translate model outputs into localized, farmer-friendly insights.

Skills & Interests

Programming & Tools

Languages: Python (Advanced), C/C++, SQL (PostgreSQL/PostGIS), JavaScript.
DevOps & Cloud: Docker, Kubernetes, GCP, CI/CD (Skaffold), FastAPI/Flask, Node.js/Express, React/Next.js, RabbitMQ, Redis.
Tools: Git, GDAL, QGIS, Google Earth Engine, Leaflet.js, LaTeX.

AI & Data Engineering

Deep Learning: PyTorch, TensorFlow, Computer Vision (OpenCV), CNN-LSTM Architectures.
Advanced AI: Self-Supervised/Contrastive Learning, RAG-based LLM Pipelines, Bayesian Modeling.
Data Science: Spatiotemporal Data Fusion, Time-Series Modeling, Multimodal Feature Engineering.
Geospatial & RS: Satellite & UAV Imagery Analysis (SAR & Multispectral), Dasymetric Modeling, Spatiotemporal Modeling.

Certifications

Data Science Professional Certificate (IBM) | C++ Essentials, CPA (Cisco) | Technical Support Fundamentals (Google) | SQL (Univ. of Michigan)